

Sagargouda Patil

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Professional Summary

Electronics and Communication Engineering graduate with a strong academic record and hands-on experience in RTL design, digital verification, FPGA implementation, and hardware validation. During my internship at DRDO, I worked on RTL verification, testbench development, UART-based communication systems, and FPGA/CPLD implementation. Through Design Verification training at Capgemini Engineering, I gained practical exposure to Verilog, SystemVerilog, simulation, debugging, and ASIC design concepts. Passionate about Design Verification, Formal Verification, and digital system development.

Education

New Horizon College of Engineering, Bangalore

2021 – 2025

Bachelor of Engineering, Electronics and Communication Engineering

CGPA: 9.56/10

Technical Skills

HDLs Verilog, SystemVerilog, VHDL

Verification RTL Verification, Functional Verification, Testbench Development, FSM Verification, Debugging, Waveform Analysis

Digital Design FPGA Design, ASIC Design Flow, Logic Synthesis, Static Timing Analysis (STA)

Protocols UART, SPI, I2C

Tools ModelSim, Vivado, Quartus Prime, Xilinx ISE, Linux

Programming C, Python, MATLAB

Experience

Capgemini Engineering

Oct 2025 – Apr 2026

Digital Design & Verification Trainee

- Worked on RTL design and verification exercises using Verilog and SystemVerilog.
- Developed testbenches and performed simulations using ModelSim.
- Debugged RTL functionality through waveform analysis and simulation results.
- Practiced FSM verification and gained exposure to ASIC design flow concepts.
- Studied synthesis fundamentals and Static Timing Analysis (STA).

DRDO – Aeronautical Development Establishment

Sep 2023 – Nov 2023

RTL Verification Intern

- Analyzed VHDL modules used in airborne encoder-decoder communication systems.
- Developed Verilog testbenches for RTL verification and functional validation.
- Simulated and verified UART communication modules using ModelSim.
- Performed debugging and waveform analysis to identify design issues.
- Implemented and validated digital designs on FPGA/CPLD platforms using Xilinx tools.

Projects

Hand Gesture Controlled Presentation System

- Developed a real-time presentation control system using Python, OpenCV, MediaPipe, and CVZone.
- Implemented hand tracking and gesture recognition for slide navigation, zooming, and presentation control.
- Improved user interaction by eliminating dependence on conventional input devices.

Muscle Controlled Robotic Hand

- Designed an EMG-based robotic hand using Arduino and biosignal sensors.
- Processed muscle activity signals to control servo motor movement.
- Demonstrated assistive technology applications for accessibility and communication.

Achievements

- Graduated with a CGPA of **9.56/10** in Electronics and Communication Engineering.
- Completed a 6-month Digital Design and Verification training program at Capgemini Engineering.
- Selected for internship at DRDO – Aeronautical Development Establishment (ADE).
- Developed hands-on expertise in RTL verification, FPGA implementation, and digital system validation.

Leadership & Activities

- Secretary, Alumni Club NHCE (2024); Treasurer and Board Member, NSS.
- NCC Cadet (Corporal); Team Lead for Sur-Sangam 2024 and Aqua Rush, Quantum X 2024.

Certifications

- Verilog HDL for Digital Design and Verification
- Digital Electronics – Complete Course (91.5 Hours)
- IBM Linux Commands and Shell Scripting